

System Architecture Documentation

This document describes key components of the system architecture used in a modern software deployment pipeline.

1. Deployment Strategy:

- The system uses canary releases to minimize deployment risk.
- Canary releases allow partial rollout of new versions to a subset of users.

2. Testing Methodologies:

- Testing includes unit tests, integration tests, and contract tests.
- Contract tests ensure that microservices adhere to agreed-upon API specifications.

3. Release Pipeline:

- CI/CD pipelines run automated tests on each merge to the main branch.
- Static analysis and security scans are part of the release process.

4. Microservices Overview:

- Authentication, billing, and reporting are separate services.
- Services communicate over REST APIs with JSON payloads.

5. Quality Controls:

- Regression testing ensures consistency of system behavior.
- Snapshot tests detect unexpected output changes.

End of sample documentation.